Digitalizing Genco O&M

Practical Use Cases and Insights

- SJSAS Smart Plant
- JEP Boiler Anomaly Detection



Presenter

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Advanced Data Analytics

Session

11:00 · Paper 1

The data already exists.
Turning it into a decision is the hard part.

01 · SCATTERED

Data in silos

- PI historian (process tags)
- GOMS (planned & unplanned outages)
- Coal certificate database
- Stockyard, UOR & correction-factor Excel files



02 · MANUAL

Stitched by hand

- Daily reports compiled every morning
- GOMS vs station records drift apart
- Heat-rate deviations explained after the fact
- Coal blends judged by scanning big tables



03 · REACTIVE

One tag at a time

- Alarms use fixed, single-tag limits
- Mill choking, slagging and fouling show up across many sensors first
- By the time one tag trips, the event has already started



Broad and deep

USE CASE A · BROAD

SJSAS Smart Plant

Manjung (TNBJ) · GF1 (M1–M3), GF2 (M4), M5

One plant-wide O&M portal: daily performance, reliability, coal & fuel advisors and an AI assistant - behind a single TNB sign-in.

- Heat rate
- UOR / EUOF
- Outage risk
- Coal blend
- Mill fineness
- ADAbot



Data PI · GOMS · coal certs · Excel
Stack React · FastAPI · PostgreSQL · LLM

USE CASE B · DEEP

JEP Boiler Anomaly Detection

Jimah East Power · Units 10 & 20

Real-time combustion monitoring: ~141 PI tags through a 3-layer AI model that flags anomalies and explains them in engineering language.

- Regime envelopes
- Isolation Forest
- SHAP
- Residual drift



Data PI Web API · 5-min · hourly sync
Stack React · FastAPI · scikit-learn · SHAP

One portal, the whole plant

SJSAS Smart Plant



PERFORMANCE

Daily Summary · Trends · Expected heat-rate model

RELIABILITY

UOR & EUOF · Outage risk · Failure modes · Reconciliation

FUEL

Coal database · Blend Advisor · M5 Fineness Advisor · Stock yard · Demurrage

AI

ADAbot chat · Daily Brief · Telegram bot

Sign in once with your **TNB AD account**. Password is checked against AD and never stored.

Four jobs, one sign-in

PERFORMANCE

- Daily heat rate & efficiency per unit, ready by 09:22
- Expected heat-rate model flags days load & coal can't explain



RELIABILITY

- UOR / EUOF + GOMS vs station reconciliation
- Monte Carlo UOR forecast: P10 / P50 / P90, next 365 days



FUEL

- Blend Advisor ranks every coal pair & ratio vs boiler limits
- M5 Fineness Advisor suggests mill settings (advisory)



AI · ADABOT

- Ask the data in plain language - web & Telegram
- **The LLM never computes a number** - the backend does



Catch boiler faults as a pattern, not an alarm

- Units 10 & 20
- ~141 PI tags
- 5-min data · hourly sync

L1

Normal for this load?

Each sensor vs its envelope in 9 MW regimes.

Catches → single-sensor excursions



L2

Still moving together?

Isolation Forest per subsystem: metal temperature, steam, fuel mill, air draft, thermal.

Catches → broken relationships



L3

Drifting slowly?

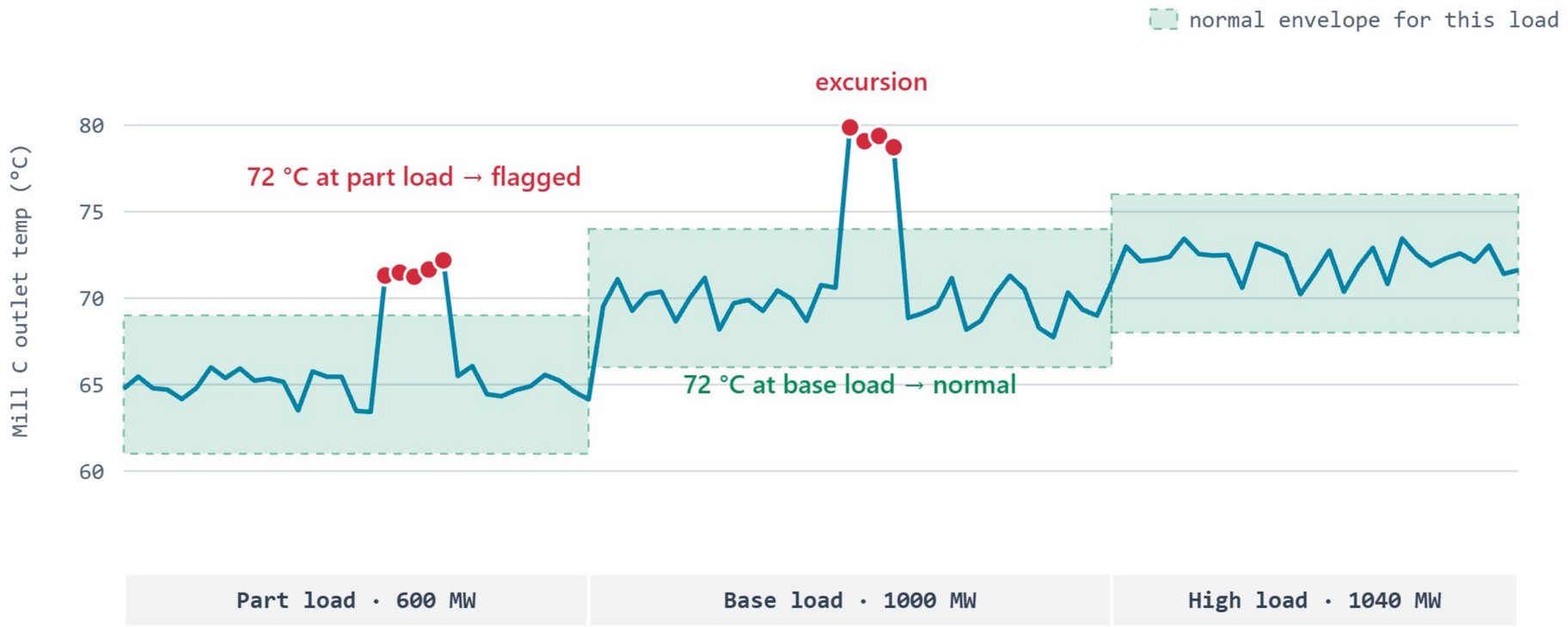
Expected output vs actual; a growing residual = degradation.

Catches → fouling, calibration drift



JEP · EXAMPLE

L1 · Normal for this load?

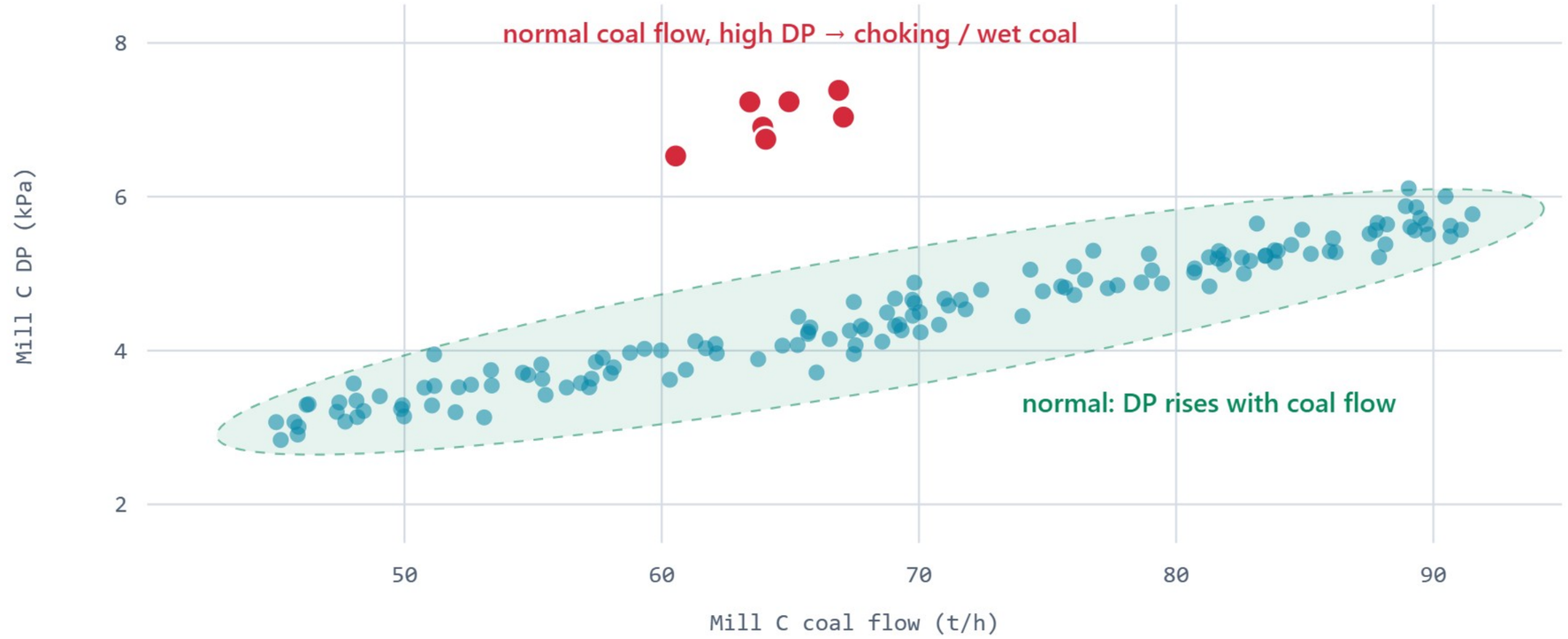


Same reading, different verdict: **72 °C is normal at base load but abnormal at part load**. L1 checks every tag against the envelope for the unit's current MW regime.

ILLUSTRATIVE EXAMPLE · NOT REAL PLANT DATA

JEP · EXAMPLE

L2 · Still moving together?



Each dot is one 5-minute snapshot of Mill C. Normal operation follows the trend. The red dots have **normal coal flow but high mill DP** - choking or wet coal - and no single tag crossed an alarm limit.

ILLUSTRATIVE EXAMPLE · NOT REAL PLANT DATA

JEP · EXAMPLE

L3 · Drifting slowly?



The model's expected output vs actual. The gap grows for weeks, too slowly to notice day to day, until the **residual crosses −2σ and is flagged** - typical of fouling or calibration drift.

ILLUSTRATIVE EXAMPLE · NOT REAL PLANT DATA

From a score to a sentence an operator can act on

ANOMALY · UNIT 20 · FUEL MILL · LAYER 2

ANOMALY

Top contributing sensors (SHAP)



"High **MILL C DP** strongly suggests a mill blockage, choking, or poor coal quality (high moisture/ash)."

ILLUSTRATIVE EXAMPLE

INSIGHT ENGINE

Turns each SHAP contributor + direction (high / low) + subsystem into plain engineering text - mills, coal flow, burners, SH / RH / economiser, furnace.

BUILT FOR THE CONTROL ROOM

- Tap a sensor → jumps to its highlighted trend
- MW load context on every chart
- Live anomaly feed per unit

INSIGHTS

What we learned

01 Start from trusted data



Numbers people already recognise get adopted.

IN PRACTICE · SJSAS

The 09:22 PI pipeline stays untouched; GOMS is reconciled against the station's own UOR file.

02 Explain or it won't be used



A named sensor and a reason, not just a score.

IN PRACTICE · JEP

"High Mill C DP → blockage, choking or wet coal" instead of an anomaly score.

03 LLM writes words, not numbers



The backend computes every figure.

IN PRACTICE · ADABOT

Plan the tools → backend computes → the model only phrases the answer.

04 Engineer in the loop



Advisors suggest; engineers decide.

IN PRACTICE · M5 ADVISOR

Suggests mill settings only inside the trained range, and says when a target is out of reach.

NEXT · 15 MINUTES

Let's see it live.

Demo, then Q&A

SJSAS SMART PLANT

3D landing · Daily Summary · Fuel advisors · ADAbot

JEP BOILER ANOMALY DETECTION

Anomaly feed · SHAP insight · sensor trend

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ADVANCED DATA ANALYTICS **genco.tnb.com.my/ada**